Steps to create wit workspace

1. Goto build portal.
2. Ask the INSU-Release Bot (This bot calls build MCP in backend)
    a. I want to create a wit workspace for branch nx_dev and tag COV..._232. Call it nx_dev_232
    b. It will generate a command to create wit workspace. (eg : /router/bin/wit space create – space nx_dev_232 –branch nx_dev –team nxos –snapshot COV..._232)
3. Paste this command of ADS machine. It will create a user_id/witspace/...
4. There will be standalone directory inside wit workspace (user_id/witspace/nx_dev_232/standalone). We can write git commands in there. (eg: git status, git diff)

Bazel packages

1. Follow steps to create wit workspace
2. Goto : user_id/witspace/nx_dev_232/standalone/bzl-packages/core_64_n9000/BUILD
3. BUILD: Gives what is needed for creating this package- Things that gets built. If bazel doesn't find some of these, it will create it.
4. BUILD file: A successful build has all of its dependencies explicitly stated in the `BUILD` file. Bazel uses those statements to create the project's dependency graph, which enables accurate incremental builds

Dependency Graph

1. Command

```
<bash > bazel --output_user_root=../bazel_cache1 build //bzl-
packages/core_64_n9000:compile_info_json_deps_graph
Extracting Bazel installation...
Starting local Bazel server (8.5.1) and connecting to it...
INFO: Analyzed target //bzl-
packages/core_64_n9000:compile_info_json_deps_graph (2677 packages loaded,
7 targets configured).
INFO: Found 1 target...
Target //bzl-packages/core_64_n9000:compile_info_json_deps_graph up-to-
date:
  bazel-bin/bzl-packages/core_64_n9000/compile_info_json_deps_graph
```

```
INFO: Elapsed time: 18.619s, Critical Path: 0.03s
INFO: 2 processes: 2 internal.
INFO: Build completed successfully, 2 total actions
```

2. `bazel-bin/bzl-packages/core_64_n9000/compile_info_json_deps_graph` is a dot structure.

3. Shows how files/libraries are connected.
4. A dependency graph per Bazel package is a directed acyclic graph (DAG) representing the relationships between build targets within a specific directory (package) and their dependencies, mapping which targets need other targets to build or execute. It acts as a blueprint of BUILD files, enabling Bazel to compute a precise, incremental build plan, ensuring only necessary components are recompiled.
5. A dependency graph is used to determine the order in which different parts of a project should be built. Having access to this allows building only what is necessary and in the correct sequence
6. https://bazel.build/concepts/dependencies

Call Graph

1. Someone is already building a call graph

Identify bazel affected targets

1. For the whattobuild Srini was mentioning this is what can be used in bazel - https://gh-nx.scm.engit.cisco.com/nx/nx/blob/nx_dev/standalone/tools/bazel_affected_targets.py
   Based on the files modified in a workspace it will find out which build targets need to be built

2. Command :

Justin: Hi, for the whattobuild Srini was mentioning this is what can be used in bazel
https://gh-nx.scm.engit.cisco.com/nx/nx/blob/nx_dev/standalone/tools/bazel_affected_targets.py
Based on the files modified in a workspace it will find out which build targets need to be built

Example:
[justijos@sjc-ads-6791 standalone]$ time ./tools/bazel_affected_targets.py --git-changed --verbose
Running command: git diff --cached --name-only

Running command: git diff --name-only
Found 2 changed files:
converged/feature/app-hosting/appmgr/appmgr_dme_handler.c
converged/feature/app-hosting/appmgr/nxsecure_evtlog.c

Finding targets that directly depend on changed files...
Analyzing file: converged/feature/app-hosting/appmgr/nxsecure_evtlog.c
Found package for converged/feature/app-hosting/appmgr/nxsecure_evtlog.c:
//converged/feature/app-hosting/appmgr
Trying query 1: attr(srcs, "nxsecure_evtlog.c", //converged/feature/app-hosting/appmgr:*)*
Running command: /auto/Insieme-EOR/eor-sw/tools/eortools/bazel --output_user_root=../bazel_cache query attr(srcs, "nxsecure_evtlog.c", //converged/feature/app-hosting/appmgr:) --output=label
Raw targets found: ['//converged/feature/app-hosting/appmgr:appmgr.bin']
converged/feature/app-hosting/appmgr/nxsecure_evtlog.c affects targets: //converged/feature/app-hosting/appmgr:appmgr.bin
Analyzing file: converged/feature/app-hosting/appmgr/appmgr_dme_handler.c
Found package for converged/feature/app-hosting/appmgr/appmgr_dme_handler.c:
//converged/feature/app-hosting/appmgr
Trying query 1: attr(srcs, "appmgr_dme_handler.c", //converged/feature/app-hosting/appmgr:*)*
Running command: /auto/Insieme-EOR/eor-sw/tools/eortools/bazel --output_user_root=../bazel_cache query attr(srcs, "appmgr_dme_handler.c", //converged/feature/app-hosting/appmgr:) --output=label
Raw targets found: ['//converged/feature/app-hosting/appmgr:appmgr.bin']
converged/feature/app-hosting/appmgr/appmgr_dme_handler.c affects targets:
//converged/feature/app-hosting/appmgr:appmgr.bin
Found 1 targets with direct file dependencies:
//converged/feature/app-hosting/appmgr:appmgr.bin

Skipping package analysis - direct dependencies found precise targets.

Found 1 affected targets:
//converged/feature/app-hosting/appmgr:appmgr.bin

real 0m4.726s
user 0m0.221s
sys 0m0.841s

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Call DB

Created a call graph db here
/auto/wsjustijos-sjc/callgraph/callgraph_vCOV_10_7_0_IDV9_0_229

You can run these commands on ads machine.

You should be able to run
export NXOS_CALLGRAPH_DB=/auto/wsjustijos-sjc/callgraph/callgraph_vCOV_10_7_0_IDV9_0_229
and then you won't have to add db to the commands.
For me that part is not working and I run it like so I add the argument
stitcher file --path utils/fsmutils/fsm_tca.c --db /auto/wsjustijos-sjc/callgraph/callgraph_vCOV_10_7_0_IDV9_0_229

Querying the Baseline
All commands below read NXOS_CALLGRAPH_DB. Add --json to any command for machine-readable output.

Search for a function

Search for a function

```
stitcher search bgp_open
stitcher search "std::vector" --limit 20
```

Function or file info

```
stitcher info --func bgp_open_error              # definition + configs
stitcher info --path routing-sw/routing/bgp/bgp_peer.c
```

Direct callees / callers

```
stitcher callees --func bgp_open_error
stitcher callers --func bgp_open_error
```

Full call chain

```
stitcher chain --func bgp_open_error --direction down --depth 3
stitcher chain --func malloc          --direction up   --depth 3
stitcher chain --func bgp_open_error --direction both --depth 2
```

All functions in a file

```
stitcher file --path utils/fsmutils/fsm_tca.c
stitcher file --path utils/fsmutils/fsm_tca.c --depth 2   # include callee trees
```

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Commit Location inside build

1. Commits available at <ins-bld-tools path for target build>/diffs/

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Problem: Identify the commit which caused build failure.

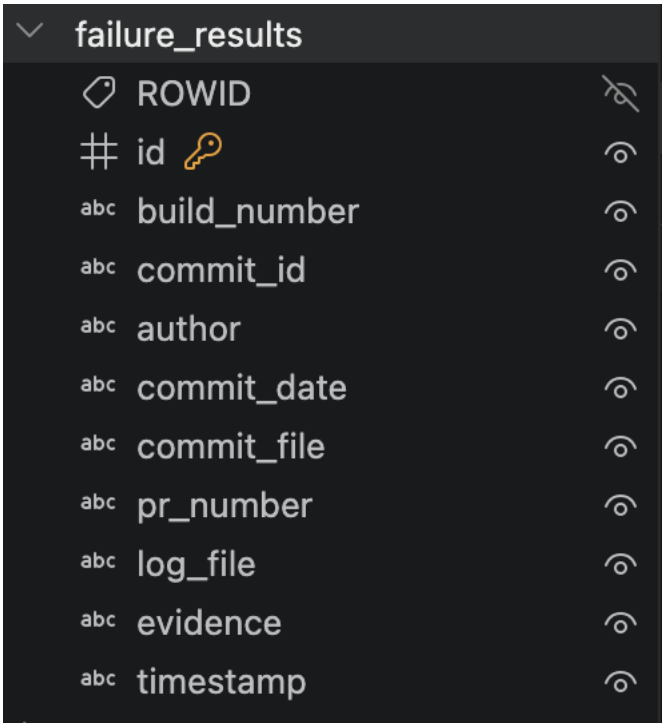
CD build # 201

```
COV_10_7_0_IDV9_0_201
  diffs
    nx_5efb3c5.diff
    nx_13559a0.diff
    nx_972300f.diff
    nx_a7a6862.diff
    nx_b0da95a.diff
    nx_COV_10_7_0_IDV9_0_200-COV_10_7_0_IDV9_0_201_diff.txt
    nx_e0076ce.diff
  .prune.error.^log.images^final^bazel^linecardimages
  .prune.error.^log.images^final^bazel^nxos64
  .prune.error.^log.images^final^bazel^nxos64_msx
  .prune.error.^log.images^final^bazel^nxos64_noncs
  .prune.error.^log.images^final^bazel^nxos64_s1
  .prune.error.^log.images^final^bazel^nxos64_s1_dpu
  .prune.error.^log.images^final^nxos64_imaging
  .prune.error.^log.images^final^nxos64_parallel
  .prune.error.^log.images^final^nxos64v_parallel
  .prune.error.^log.images^final^nxosvp
```

- 6 commits in 1 build
- Problem to solve : Based on error logs, identify which commit caused the build failure.
- Output:

- ```
namane@NAMANE-M-3YKQ workspace % python3 commit_dep/diffs/find_failure_commits_db.py commit_dep/.prune.error.^log.images^final^bazel^linecardimages
      "commit_date": "Mon Apr 6 15:08:34 2026 -0400",
      "commit_file": "nx_13559a0.diff",
      "commit_id": "13559a0952b8f217fb608d89af2c5aad79cabf30",
      "evidence": [
        "exact file match: standalone/platform/insieme/drivers/cctrli/BUILD",
        "same directory touched: standalone/platform/insieme/drivers/cctrli",
        "same directory touched: standalone/platform/insieme/drivers/cctrli/platform/tor-wistron"
      ],
      "log_file": ".prune.error.^log.images^final^bazel^linecardimages",
      "pr_number": "86527",
      "timestamp": "2026-04-24T04:32:07.117835+00:00"
    }
  ],
  "timestamp": "2026-04-24T04:32:07.117835+00:00"
}
namane@NAMANE-M-3YKQ workspace %
```
- 

| id | build_num... | commit_id | author | commit_date | commit_... | pr_num... | log_file | evidence | timestamp |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 201 | 13559a0952b8... | chcheste | Mon Apr 6 15:08:34... | nx_13559a0.diff | 86527 | .prune.error.^l... | ["exact file ... | 2026-04-23T22:56:11.899206+00:00 |
| 2 | 201 | 13559a0952b8... | chcheste | Mon Apr 6 15:08:34... | nx_13559a0.diff | 86527 | .prune.error.^l... | ["exact file ... | 2026-04-24T04:32:07.117835+00:00 |

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**Identify the consecutive build that failed**

**Set of consecutive build failures.**

1. 101 – 104
```

a.
```
./COV_10_7_0_IDV9_0_101/
./COV_10_7_0_IDV9_0_101/
./COV_10_7_0_IDV9_0_102/
./COV_10_7_0_IDV9_0_102/
./COV_10_7_0_IDV9_0_102/
./COV_10_7_0_IDV9_0_103/
./COV_10_7_0_IDV9_0_104/
```

2. 114 –116
3. 121- 124
4. 127 – 130
5. 140 – 141
6. 146 – 147
7. 153 - 154
8. 198 – 199 (COV_10_7_0_IDV9_0_198 -COV_10_7_0_IDV9_0_199)
9. 159 – 163
10. 168 – 169
11. 182 - 184

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